

# DP Computer Science: A 2.2.1 Network Topologies

## 1. Core Concepts & Learning Objectives

### Learning Intentions

By the end of this lesson, students will be able to:

- Define and describe different network topologies: **bus, ring, star, tree, mesh, hybrid**
- Analyse the **advantages and disadvantages** of each topology
- Evaluate the **factors influencing network design** (cost, scalability, reliability, etc.)
- Apply knowledge to **recommend suitable topologies** for different settings
- Explain **emerging trends** in network architecture (SDN, cloud computing)

### Key Vocabulary

| Term             | Definition                                                            |
|------------------|-----------------------------------------------------------------------|
| Network topology | The physical or logical arrangement of devices in a network           |
| Bus topology     | All devices connect to a single communication line (cable)            |
| Ring topology    | Devices are connected in a closed loop/circle                         |
| Star topology    | All devices connect to a central hub or switch                        |
| Tree topology    | A hierarchical structure with a root node and branches                |
| Mesh topology    | Every device connects to every other device (fully connected)         |
| Hybrid topology  | A combination of two or more topologies                               |
| Node             | Any device connected to the network (computer, printer, switch, etc.) |
| Fault tolerance  | The ability of a network to continue functioning when parts fail      |
| Scalability      | The ability of a network to grow and accommodate more devices         |

### IB ATL Skills

| Skill           | Description                                                               |
|-----------------|---------------------------------------------------------------------------|
| Thinking        | Critical thinking, analysis, problem-solving                              |
| Communication   | Explaining concepts, presenting information, collaborating in discussions |
| Research        | Gathering information and evaluating sources                              |
| Self-Management | Organisation, time management, independent learning                       |

## 2. Network Topologies Overview

## What is a Network Topology?

Network topology describes how devices connect in a network—the physical or logical arrangement of nodes and links.

The choice of topology affects:

- **Cost** (cabling, hardware)
- **Reliability** (fault tolerance)
- **Performance** (speed, collisions)
- **Scalability** (ease of adding devices)
- **Complexity** (installation and maintenance)

### Topology Comparison at a Glance

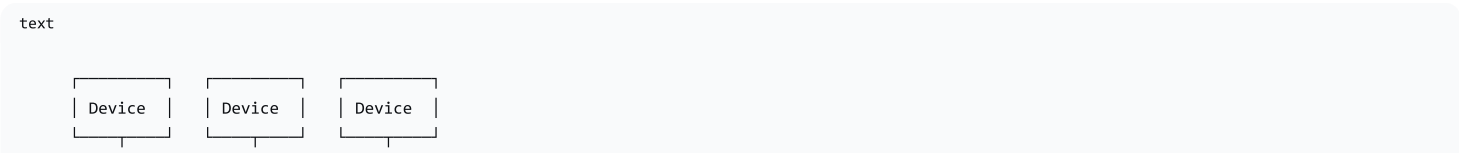
| Topology | Diagram                                                                           | Cost     | Reliability          | Scalability | Best For                                      |
|----------|-----------------------------------------------------------------------------------|----------|----------------------|-------------|-----------------------------------------------|
| Bus      |  | Low      | Poor                 | Limited     | Small temporary networks                      |
| Ring     |  | Medium   | Medium               | Limited     | Small to medium networks                      |
| Star     |  | Low      | Good (hub-dependent) | Good        | Homes, small offices                          |
| Tree     | Hierarchical                                                                      | Medium   | Good                 | Excellent   | Large organisations, campuses                 |
| Mesh     | Fully connected                                                                   | High     | Excellent            | Good        | Critical systems, military, internet backbone |
| Hybrid   | Combination                                                                       | Variable | Variable             | Excellent   | Large, complex networks                       |

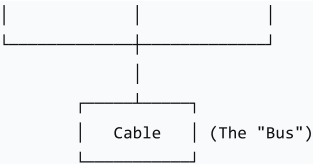
## 3. Detailed Topology Descriptions

### Bus Topology

| Aspect         | Description                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------|
| Structure      | All devices connect to a single cable (the "bus")                                                                      |
| How it works   | Data travels along the cable; all devices see the data but only the intended recipient processes it                    |
| Advantages     | Simple, inexpensive, easy to set up, requires less cable than other topologies                                         |
| Disadvantages  | Single point of failure (the cable); limited cable length; collisions slow down the network; difficult to troubleshoot |
| Scalability    | Poor—adding devices slows the network                                                                                  |
| Real-world use | Small temporary networks; legacy networks; simple setups                                                               |

Diagram:



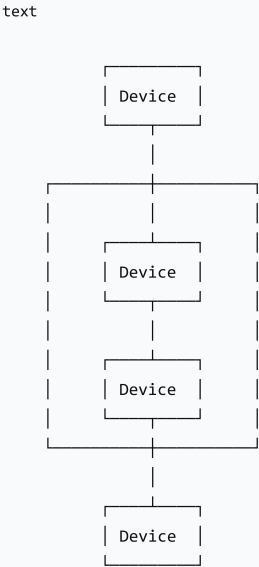


**Analogy:** A bus topology is like a **shared hallway**—everyone shares the same space, and messages are passed along until they reach the right person.

## Ring Topology

| Aspect         | Description                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------|
| Structure      | Devices are connected in a closed loop/circle                                                                      |
| How it works   | Data travels in one direction (or both) around the ring; each device passes data to the next                       |
| Advantages     | Ordered transmission (no collisions); predictable data flow; simple cabling                                        |
| Disadvantages  | One failed device can break the whole network; difficult to troubleshoot; adding devices requires network downtime |
| Scalability    | Limited—large rings can become slow                                                                                |
| Real-world use | Small to medium networks; legacy networks (Token Ring); SONET/SDH networks                                         |

Diagram:



**Analogy:** A ring topology is like a **round-robin meeting**—each person gets a turn to speak, and messages are passed around the circle.

## Star Topology

| Aspect         | Description                                                                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Structure      | All devices connect to a central hub or switch                                                                                                            |
| How it works   | Data travels from device → central hub → destination device                                                                                               |
| Advantages     | Easy to install and manage; fault-tolerant (one cable failure doesn't affect others); easy to add devices; good performance (no collisions with switches) |
| Disadvantages  | Central hub is a single point of failure; requires more cabling than bus                                                                                  |
| Scalability    | Good—add devices by connecting to the hub                                                                                                                 |
| Real-world use | Home networks, small offices, school computer labs                                                                                                        |

Diagram:



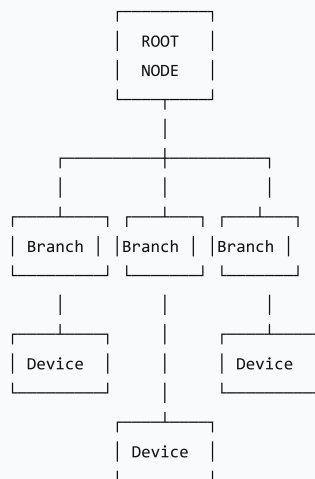
Analogy: A star topology is like a **school office**—all messages go through the office, which distributes them to the correct person.

## Tree Topology

| Aspect         | Description                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------|
| Structure      | A hierarchical arrangement with a root node and branches (combines star and bus)                |
| How it works   | Data travels from leaf nodes up through the hierarchy to the root, then down to other branches  |
| Advantages     | Scalable; hierarchical management; good performance; fault-tolerant if designed with redundancy |
| Disadvantages  | Root node is critical; more complex to design; more cabling than simple topologies              |
| Scalability    | Excellent—add branches as needed                                                                |
| Real-world use | Large organisations, university campuses, corporate networks                                    |

Diagram:

text



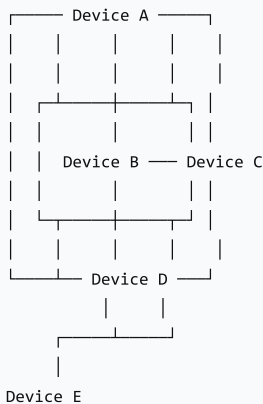
**Analogy:** A tree topology is like a **company org chart**—information flows from the CEO down through departments to individual employees.

## Mesh Topology

| Aspect         | Description                                                                                |
|----------------|--------------------------------------------------------------------------------------------|
| Structure      | Every device connects to every other device (fully connected)                              |
| How it works   | Multiple paths between any two devices; data can choose alternative routes                 |
| Advantages     | Extremely reliable; fault-tolerant (multiple paths); redundant; no single point of failure |
| Disadvantages  | Expensive; complex to install; requires significant cabling; difficult to manage           |
| Scalability    | Limited—each new device requires connections to all others $(n(n-1)/2)$ connections)       |
| Real-world use | Military communication systems; critical infrastructure; internet backbone                 |

**Diagram (Full Mesh with 5 devices):**

text





**Analogy:** A mesh topology is like a **city road network**—there are multiple routes to get from one place to another, so if one road is blocked, you can take another.

## Hybrid Topology

| Aspect         | Description                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------|
| Structure      | A combination of two or more topologies                                                          |
| How it works   | Different parts of the network use different topologies based on their needs                     |
| Advantages     | Flexible; customisable; can meet diverse requirements; combines strengths of multiple topologies |
| Disadvantages  | Complex to design; expensive; difficult to manage                                                |
| Scalability    | Excellent—different areas can be scaled independently                                            |
| Real-world use | Large enterprise networks; university campuses; ISP networks                                     |

## 4. Factors Influencing Topology Choice

| Factor            | Description                                | How It Affects Choice                                          |
|-------------------|--------------------------------------------|----------------------------------------------------------------|
| Cost              | Budget for hardware, cabling, installation | Lower cost → bus or star; higher cost → mesh or hybrid         |
| Reliability       | Importance of network uptime               | High reliability → mesh or star (with redundant hub)           |
| Scalability       | Need to add more devices                   | High scalability → tree, star, or hybrid                       |
| Speed/Performance | Data transfer requirements                 | Star or mesh (switches reduce collisions)                      |
| Complexity        | Ease of installation and management        | Low complexity → star or bus; high complexity → mesh or hybrid |
| Physical Layout   | Building size, geography                   | Large area → tree or hybrid; small area → star                 |
| Fault Tolerance   | Ability to survive failures                | High fault tolerance → mesh or ring (with dual-ring)           |

## 5. Topologies in Different Settings

| Setting      | Recommended Topology | Why?                                                              |
|--------------|----------------------|-------------------------------------------------------------------|
| Home Network | Star                 | Easy to set up, affordable, suitable for small numbers of devices |

| Setting                   | Recommended Topology | Why?                                                                           |
|---------------------------|----------------------|--------------------------------------------------------------------------------|
| Small Office              | Star or Hybrid       | Easy to manage; can add devices easily; can add backup hub for redundancy      |
| University Campus         | Hybrid (Star + Tree) | Connects multiple buildings and departments; scalable; hierarchical management |
| Military Communication    | Mesh                 | Highest reliability and fault tolerance; survives damage                       |
| Airplane Internal Network | Hybrid               | Critical systems need redundancy (mesh); entertainment may use star/bus        |
| Internet Backbone         | Mesh                 | High redundancy; handles massive traffic; multiple paths                       |
| Smart Home                | Mesh (wireless)      | Reliable coverage throughout the house; self-healing                           |
| Rural Community Wi-Fi     | Mesh (wireless)      | Extends network range; shares internet connection across households            |

## 6. Emerging Trends in Network Architecture

| Trend                             | Description                                                                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------------------|
| SDN (Software-Defined Networking) | Separates network control from data forwarding; makes networks more flexible and programmable           |
| Cloud Computing                   | Networks increasingly rely on cloud services; topology must support cloud connectivity                  |
| Wireless Mesh Networks            | Self-configuring, self-healing wireless networks; growing in popularity for smart homes and communities |
| IoT (Internet of Things)          | Massive numbers of devices connected; requires scalable, efficient topologies                           |
| 5G Networks                       | High-speed, low-latency mobile networks; uses complex hybrid topologies                                 |

## 7. Lesson Sequence

### Lesson Plan

| Phase  | Time   | Activity                                                                |
|--------|--------|-------------------------------------------------------------------------|
| Engage | 10 min | "How does your phone connect to the internet?"; define network topology |

| Phase                | Time   | Activity                                                         |
|----------------------|--------|------------------------------------------------------------------|
| Interactive Activity | 15 min | Think-Pair-Share: Match scenarios to topologies; justify choices |
| Explain              | 15 min | Core topologies (bus, star, mesh); factors influencing design    |
| Elaborate            | 15 min | Additional topologies (ring, tree, hybrid); emerging trends      |
| Wrap-up              | 5 min  | Key takeaways; exit ticket                                       |

Activity: Network Topology Match-Up

Instructions: Students match each scenario to the most suitable topology and justify their choice.

| Scenario                  | Best Topology        | Justification                                                  |
|---------------------------|----------------------|----------------------------------------------------------------|
| Home Network              | Star                 | Easy to set up, affordable, suitable for small networks        |
| Small Office              | Star or Hybrid       | Reliable; easy to manage; can add devices as needed            |
| University Campus         | Hybrid (Star + Tree) | Connects multiple buildings; scalable; hierarchical management |
| Military Communication    | Mesh                 | Highest reliability and fault tolerance                        |
| Airplane Internal Network | Hybrid               | Critical systems need redundancy; entertainment less critical  |
| Internet Backbone         | Mesh                 | Extremely robust; handles massive traffic                      |
| Smart Home                | Mesh                 | Reliable coverage; self-healing                                |
| Rural Community Wi-Fi     | Mesh                 | Extends range; shares internet connection                      |

8. Assessment Activities

| Assessment Type           | Description                                                         |
|---------------------------|---------------------------------------------------------------------|
| Class Participation       | Observe engagement and contributions                                |
| Network Topology Match-Up | Assess ability to match scenarios to topologies and justify choices |
| Exit Ticket               | Gauge understanding and identify areas needing clarification        |

Sample Exit Ticket Questions:

- 1. What is a network topology?
- 2. What is a disadvantage of a bus topology?
- 3. Why is a mesh topology used for critical systems?
- 4. Which topology is most common in home networks and why?



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## 9. Differentiation

| Level     | Strategy                                                                      |
|-----------|-------------------------------------------------------------------------------|
| Support   | Provide visual aids, concise definitions, and real-world examples             |
| Challenge | Encourage advanced students to research SDN, 5G, or IoT network architectures |

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## 10. Resources

- **Presentation** (Slide Deck)
  - **eBook** (A2.2.1 with quiz)
  - **Quizlet** (A2.2.1 vocabulary flashcards)
  - **Activity:** Network Topology Match-Up handout
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## 11. Summary: Key Takeaways

| Topology | Key Feature                     | Best For                            |
|----------|---------------------------------|-------------------------------------|
| Bus      | Simple, cheap, single cable     | Small temporary networks            |
| Ring     | Closed loop, ordered data flow  | Small to medium networks            |
| Star     | Central hub, easy to manage     | Homes, small offices                |
| Tree     | Hierarchical, scalable          | Large organisations, campuses       |
| Mesh     | Fully connected, fault-tolerant | Critical systems, internet backbone |
| Hybrid   | Combination, flexible           | Large complex networks              |

| "The best topology depends on the specific needs—cost, reliability, scalability, and fault tolerance all play a role in the decision."